MA1513 Weekly Self Practice 2 Answer

1. (a) Let $\boldsymbol{A} = \begin{pmatrix} 1 & a & 0 \\ 0 & 0 & 1 \\ b & 0 & 0 \end{pmatrix}$. Here a and b represent some real numbers that are not zero. Find $\boldsymbol{A}^{-1}$ using Gauss-Jordan elimination. (Show your working.)

   (b) Use the expansion method to find the determinant of the matrix $\boldsymbol{A}$ in (a). (Show your working.)

Answer

(a)

$$\left(\begin{array}{ccc|ccc} 1 & a & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ b & 0 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 - bR_1} \left(\begin{array}{ccc|ccc} 1 & a & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & -ab & 0 & -b & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|ccc} 1 & a & 0 & 1 & 0 & 0 \\ 0 & -ab & 0 & -b & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array}\right) \xrightarrow{-\frac{1}{ab}R_2} \left(\begin{array}{ccc|ccc} 1 & a & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{a} & 0 & -\frac{1}{ab} \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array}\right)$$

$$\xrightarrow{R1 - aR_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & \frac{1}{b} \\ 0 & 1 & 0 & \frac{1}{a} & 0 & -\frac{1}{ab} \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array}\right)$$

So $\boldsymbol{A}^{-1} = \begin{pmatrix} 0 & 0 & \frac{1}{b} \\ \frac{1}{a} & 0 & -\frac{1}{ab} \\ 0 & 1 & 0 \end{pmatrix}$

(b)

$$\det \boldsymbol{A} = 1 \times \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} - a \times \begin{vmatrix} 0 & 1 \\ b & 0 \end{vmatrix} + 0 \times \begin{vmatrix} 0 & 0 \\ b & 0 \end{vmatrix} = 0 + ab + 0 = ab$$

2. (a) Convert the vector equation

$$(1,-4,4,1) = a(1,-1,2,0) + b(1,2,0,-1)$$

into a linear system with variables a and b and write down its augmented matrix.

(b) Use the `rref` command in MATLAB to find the value of a and b. (Do not perform Gaussian elimination.)

(c) Use the above result to determine whether $(1,-4,4,1)$ belongs to

$$\text{span}\{(1,-1,2,0),(1,2,0,-1)\}?$$

Give a brief explanation.

Answer

(a) The vector equation can be simplified as $(1,-4,4,1) = (a+b,-a+2b,2a,-b)$ which gives the linear system

$$\begin{array}{ccccc} a & + & b & = & 1 \\ -a & + & 2b & = & -4 \\ 2a & & & = & 4 \\ & & -b & = & 1 \end{array}.$$

(b) The augmented matrix is given by:

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ -1 & 2 & -4 \\ 2 & 0 & 4 \\ 0 & -1 & 1 \end{array}\right)$$

Applying MATLAB command `rref` to this augmented matrix gives:

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

The RREF gives the solution $a = 2, b = -1$.

(c) $(1,-4,4,1)$ belongs to $\text{span}\{(1,-1,2,0),(1,2,0,-1)\}$. This is because the system in (a) is consistent and hence $(1,-4,4,1)$ is a linear combination of $(1,-1,2,0)$ and $(1,2,0,-1)$.

3. Consider the matrices $\boldsymbol{C} = \begin{pmatrix} 12 & 34 & 56 \\ 78 & 90 & 12 \\ 34 & 56 & 78 \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} 12 & 13 & 14 \\ 22 & 23 & 24 \\ 32 & 33 & 34 \end{pmatrix}$

   (a) Use MATLAB command to find the ranks of $\boldsymbol{C}$ and $\boldsymbol{D}$.

   (b) Which of $\boldsymbol{C}$ and $\boldsymbol{D}$ is full rank?

   (c) Without further working, determine which of the homogeneous systems $\boldsymbol{Cx} = \mathbf{0}$ and $\boldsymbol{Dx} = \mathbf{0}$ has non-trivial solutions.

Answer

(a) By using either the MATLAB command $\gg$ `rank` or $\gg$ `rref`, we have $\text{rank}(\boldsymbol{C}) = 3$ and $\text{rank}(\boldsymbol{D}) = 2$.

(b) $\boldsymbol{C}$ is full rank (size 3×3 and rank 3).

(c) $\boldsymbol{Dx} = \mathbf{0}$ has non-trivial solutions, since $\boldsymbol{D}$ is not full rank, and hence is a singular matrix.